13D research: ML, NLP and long-term active risk

Investment aim: use machine learning (ML) and natural language processing (NLP) to uncover long-term alpha signals in public companies backed by high-quality event-driven or activist hedge funds, then take active risk in several selected, potentially concentrated positions.

Status: the historical simple-following baseline has not established a reliable investable edge. Filing preparation and manager scorecards are underway; predictive ML and incremental intent NLP have not been evaluated.

The intended output is a ranked set of company opportunities with an economic thesis, catalyst, downside case and position rationale. Manager reputation and a disclosed plan are candidate inputs; their value must be tested using information available before purchase.

1. Three phases - baseline, selection and disclosed intentions

Phase / purpose	Information and comparison	What has been done
1. Simple following Measure post-disclosure returns.	Initial 13D events and manager groups; stock returns versus Russell size-segment ETF proxies.	Historical baseline evaluated. No reliable edge established; some funded simulations incomplete.
2. Structured ML Rank stronger opportunities.	Investor history, ownership, company size, liquidity, volatility and prior returns; fundamentals planned.	15,813 candidates audited; HFR scorecards prepared. Dated joins/features unfinished; no fitted model.
3. Incremental NLP Add the investor's stated plan.	Item 4 objectives and specificity; compare structured ML with ML plus text on the same sample.	Planned. No validated intent/alpha result. Ownership/identity extraction is preparation work.

2. Data sources and their role

Source	Information	Use / status
SEC EDGAR	13D/13D-A documents, public acceptance times, filer/issuer identities, stakes and Item 4 text.	Cached source-audited filing archive; factual extraction and future intent NLP. [E2]
CRSP via WRDS	Daily stock/ETF prices, returns, volume, historical security IDs and delisting fields.	Market data behind the reported historical return tests and simulations. [E1, E4]
FTSE Russell via WRDS	Historical Russell constituent snapshots; ETF returns come from CRSP.	Stock-appropriate size-segment ETF benchmark assignments. [E4]
HFR	35 supplied quarterly snapshots, 2018 Q1-2026 Q3: fund metadata, strategy, reported returns and assets.	Manager candidates and descriptive peer scorecards; predictive value untested. [E3]
Compustat via WRDS	Fundamentals for valuation, leverage, profitability and balance-sheet features.	Planned features. A historically available, joined ML panel is unfinished. [E4]

WRDS = Wharton Research Data Services, the access platform for the separate CRSP, FTSE Russell and Compustat datasets.

HFR fund performance is different from target-stock returns available to an outside follower. Dated manager identities, availability and incomplete dead-fund coverage need validation before the scorecards can become signals. [E3]

Proposed extension: amendments for position adjustments

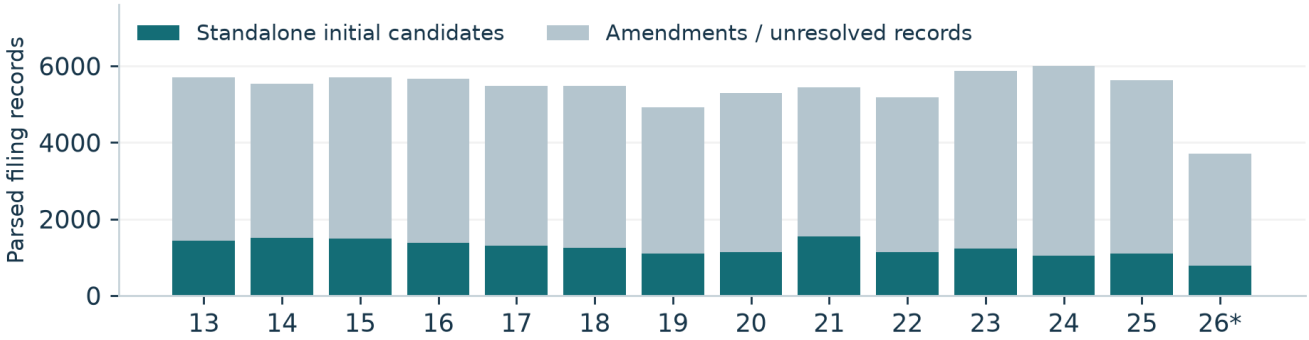
Amendment disclosure	Potential use and observability
Stake increases / reductions	Compare Item 5 shares, issuer ownership percentages and transactions. Percentage changes may reflect issuer share-count changes; ownership percentage is not fund portfolio weight.
Exit / reporting threshold	Item 5(e) dates a fall to 5% or less, which is not necessarily zero. Confirm a full disposition explicitly or leave the full-exit date unknown. [E5]

From about 75,000 filings to 15,813 candidates

Cached SEC archive: January 2, 2013-September 11, 2026 | Unit: parsed filing records

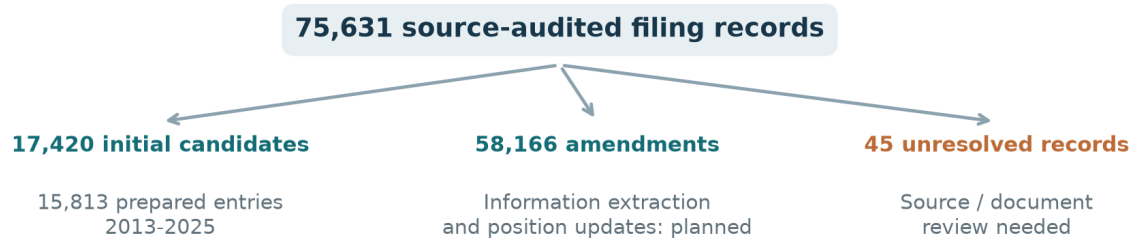
The audited archive contains **75,631 records**, including **58,166 amendments**, **17,420 standalone initial-filing candidates** and **45 unresolved records**. These are filing records, not unique companies, managers or independently validated campaigns. [E2]

3. Archive coverage by acceptance year



Years shown as 13-26. *2026 is partial through September 11. The initial-candidate segment is before the staging exclusions below. This is the project's cached, parsed archive, not a completeness claim about every SEC filing. [E2]

4. Two research streams: initial entries and amendment information



Initial-entry preparation step	Set aside	Remaining
Source-audited archive	-	75,631
Protect review records/issuers; apply other user exclusions	3,107	72,524
Restrict acceptance dates to December 31, 2025 or earlier	3,613	68,911
Remove unresolved source/primary-document cases	28	68,883
Retain initial candidates; keep other records as separate streams	53,070	**15,813**

Set aside means outside the initial-entry cohort, not discarded from the archive. Counts reconcile across the full archive in priority order. The 53,070 includes remaining amendments and other non-initial records after earlier checks; it differs from the raw 58,166 amendment count. [E2]

What the 15,813 represent: initial-filing candidates accepted in 2013-2025, all processed by the factual staging audit. They are not yet a validated hedge-fund universe or a tradable ML panel. Protected review samples remain independent.

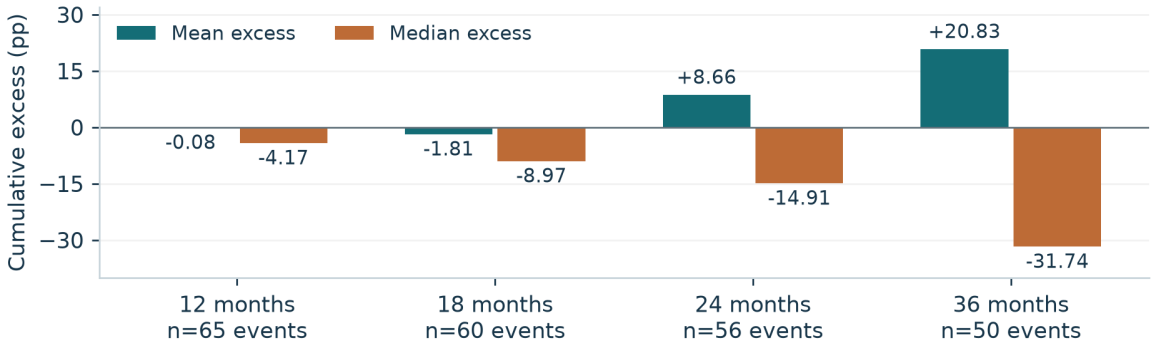
Amendment stream: retain and classify disclosures for ownership, intent, campaign and exit information (page 4). Bucket counts and a validated amendment feature panel are not yet measured. Apply the same date limits and independent-review protections before modeling.

Separate return-study sample: Phase 1 uses a 2018-2025 schedule and additional security, benchmark and outcome checks. At 24 months, the broad sample is 2,107 scheduled > 712 benchmark-assigned > 694 calculable; its focused activist-candidate subset is 67 > 58 > 56. The 56 events are not a direct funnel from the later 15,813-record staging exercise. [E1]

Historical returns: positive averages, weak medians

Phase 1 exploratory development | 2018-2025 research window | Focused activist-candidate subset

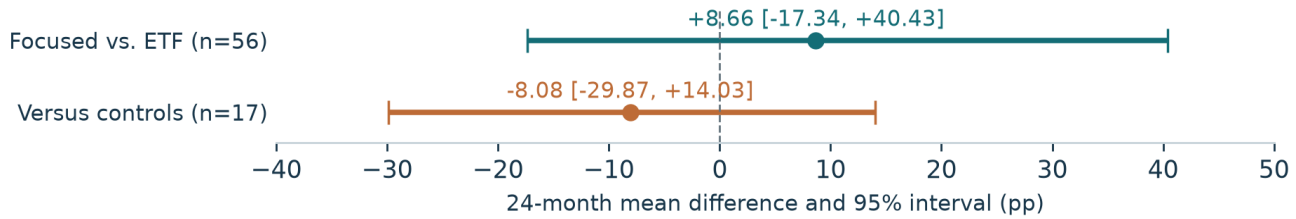
5. The mean improves with horizon; the median remains negative



Hold	n	Stock mean	ETF mean	Excess mean	Excess median	Beat ETF	Mean excess 95% interval
12 mo	65	+11.58%	+11.65%	-0.08 pp	-4.17 pp	47.7% (31/65)	-11.25 to +13.64 pp
18 mo	60	+13.52%	+15.34%	-1.81 pp	-8.97 pp	35.0% (21/60)	-19.82 to +23.45 pp
24 mo	56	+27.82%	+19.15%	+8.66 pp	-14.91 pp	41.1% (23/56)	-17.34 to +40.43 pp
36 mo	50	+53.57%	+32.74%	+20.83 pp	-31.74 pp	32.0% (16/50)	-20.87 to +84.21 pp

Gross, cumulative, equally weighted event returns; excess = stock minus its Russell size-segment ETF proxy; pp = percentage points. Horizons use different, overlapping calculable samples, so higher means do not establish an optimal holding period. Intervals use the saved issuer bootstrap. [E1]

6. Two-year results weaken under a matched comparison



Both 95% intervals cross zero. In 17 industry/year/size-matched cases, focused-event excess averaged -3.07 pp versus +5.01 pp for controls: difference -8.08 pp. This subset differs from the 56-event ETF comparison; the two estimates must not be pooled. [E1]

7. A funded account is a separate test

Completed $100k / 12-month account	Result	Comparison / context
March 19, 2018-December 31, 2025	+57.41% cumulative +5.99% annualized	Cash-only +22.87%; SPY +181.73%; IWV +166.93% cumulative.
Deployment and downside	75 buys; 55.41% average cash -35.15% maximum drawdown	5% issuer targets, max. 20; cash interest, liquidity limits and modeled costs. Different exposures prevent alpha claims.

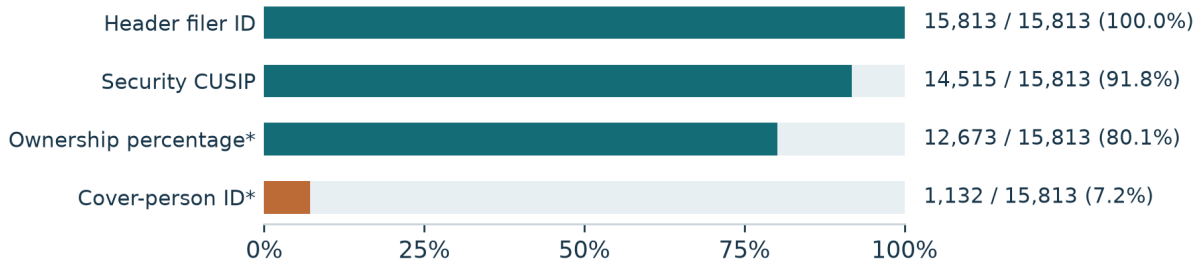
Conventions: modeled entry at the opening price on the second exchange session after SEC acceptance; 252/378/504/756 sessions approximate 12/18/24/36 months. Observed opens do not guarantee fills. No simultaneous duplicate-issuer positions in funded simulations. Historical stock/benchmark coverage is selective; monthly membership dates have availability limits.

Primary $100k 18/24/36-month accounts remain incomplete at Aratana's replacement-share/contingent-right settlement; proceeds are not invented. The gross event table does not resolve that account issue. Holding horizons were revised after inspecting earlier results; this is exploratory evidence. [E1]

From prepared data to an investable selection test

Next research steps | Candidate mechanisms, not demonstrated alpha signals

8. Extraction coverage is substantial; economic links need work



15,813 selected 2013-2025 filings. *At least one extracted value, not complete person/ownership coverage. Bars overlap. Header IDs/CUSIPs do not establish validated fund/stock links; presence does not measure extraction accuracy. [E2]

9. Proposed signals and their investment mechanism

Candidate signal	Economic hypothesis	Evidence needed
Manager quality / history Dated HFR peers; prior target-stock outcomes.	Repeated execution skill may distinguish managers whose plans create value.	Validate historical identities and availability; separate fund returns from outside-follower returns.
Company economics Valuation, leverage, cash flow, liquidity.	A credible catalyst may have more upside where value and financial capacity support it.	Build dated fundamentals/market features; test against size, industry and risk explanations.
Disclosed intent (NLP) Board change, asset sale, capital return; plan specificity.	A concrete, feasible plan may change cash flows or capital allocation beyond what prices reflect.	Validate text labels against source spans; compare ML with and without NLP on identical events.

10. Proposed amendment buckets for extracting new information

Information bucket	Candidate fields / use
Ownership change	Shares, ownership %, dated buys/sales; distinguish additions/reductions from issuer share-count changes.
Intent or campaign development	Changed Item 4 plans, board representation, settlements, asset sales or agreements; update thesis/catalyst evidence.
Exit / reporting transition	Keep confirmed full exits, 5%-or-less disclosures and reporting-regime changes distinct; unresolved full-exit dates stay unknown.
Identity / technical update	Reporting-person/group changes and corrections; improve attribution without assuming a trade.
Unclear or incomplete	Preserve evidence and missingness for review; do not infer no activity from incomplete extraction.

Proposed buckets are overlapping labels, not measured mutually exclusive counts. One amendment may change both stake and intent. Link it to prior filings, carry forward unchanged facts, and preserve source spans plus transaction/disclosure dates. This panel and full-exit labels are unfinished.

Next validation: freeze dated inputs and compare simple following, structured ML, ML plus NLP, then amendment-driven adjustments under common trading/cost rules. Trade after disclosure; test downside, liquidity and shared exposures before concentrated sizing. Reserve unexposed or prospective evaluation; inspected history is development data.

Evidence key: E1: Sept 24 long-horizon summaries, matched controls and Phase 1 conclusion. E2: universe ledger/staging selection and coverage audit. E3: Sept 30 HFR protocol/scorecards. E4: CRSP/FTSE source inventory and WRDS records. E5: SEC guidance (103.08, 104.05, 110.05/08) and 2023 final rule. Local source links in companion notes.

Prepared October 4, 2026 from preserved project evidence. AI-assisted synthesis and graphics; no new backtest, trained model, intent labels or changes to independent human-review materials.